



Multimodal Tensions and the Foregrounding of Mathematical Content: A Textbook Analysis of Introductions to Equality and Subtraction

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Abstract

This study investigates how tensions within and between modes, understood as contrasts or discontinuities in how mathematical content is communicated, are constituted in textbook introductions of equality and subtraction. While multimodal research in mathematics education has emphasized cohesion and coordination between modes, less attention has been given to discontinuities and contrasts, which may create conditions for productive struggle arising in students. Adopting a multimodal perspective grounded in the concept of tension, the study conducts an exploratory analysis of introductory chapters on equality and subtraction in three Swedish grade-one textbook series. Through a bottom-up categorization, five types of multimodal tension were identified. Three occur at a macro level, unfolding across successive parts, for example, between an example and a task, while two occur at a micro level within single parts. The findings show that different types of tension establish qualitatively different relations between modes and mathematical content. Macro-level tensions tend to foreground invariant aspects of equality or subtraction through variation over time, whereas micro-level tensions more often fragment or ambiguously anchor the mathematical content within one part of the textbook. By distinguishing between macro- and micro-level tensions and relating them to specific mathematical content, the study contributes a more fine-grained vocabulary for analysing multimodal textbook design. The results contribute to a more nuanced understanding of multimodal tension by showing that its potential to support productive struggle depends on where and how it is constituted in relation to mathematical ideas.

Keywords Cohesion and tension · Mathematical modes · Mathematical textbooks · Productive struggle

Extended author information available on the last page of the article

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Introduction

To learn and use mathematics, students need to make meaning from multiple modes, which in this study refers to socially and culturally shaped resources for meaning-making (e.g., natural language, mathematical symbols, images, diagram, and physical objects). Representations are understood as particular realizations of mathematical ideas through these modes (e.g., Duval, 2006). Thus, the same mode may be used to represent different aspects of a mathematical idea, while the same mathematical idea may be represented through several modes. From different theoretical perspectives, such as cognitive (Duval, 2006), linguistic (Schleppegrell, 2007), and discursive (O'Halloran, 2005), mathematical understanding is described as dependent on how students coordinate these modes. Particularly, mathematical learning requires students to coordinate and move between different representations of the same mathematical idea (Duval, 2006), and mathematical understanding implies an understanding of how various modes are related to each other (Wedman, 2020). Moreover, students need opportunities to encounter mathematical ideas in varied ways in order to develop flexible and robust understanding (Niss, 2006). Thus, mathematical learning is not only a matter of encountering multiple modes, but of how these modes interact in relation to specific mathematical content (i.e., the mathematical ideas that the modes collectively communicate and make available for students to discern).

At the same time, a general idea for learning mathematics is that students need to actively engage with mathematical contents that do not immediately make sense to them. For example, to be engaged in productive struggle (Hiebert & Grouws, 2007), productive failure or creative mathematical reasoning (Jonsson et al., 2014; Lithner, 2017; Norqvist et al., 2019). Common for the above constructs is that a task needs to offer some challenges to students to provide opportunity for active engagement, but at the same time not being overly difficult so students have reasonable mathematical foundation to build on. Thus, whether a task contributes to productive struggle depends on the relation between the task and the student, and what type of tasks students have worked on previously. In textbooks, this may depend on how mathematical content is sequenced and how explanations, examples, and tasks direct students' attention to particular aspects of that content. Empirical studies show that such design features influence which parts of a task students attend to, for example whether attention is directed towards provided solution procedures or towards mathematically informative representations (Norqvist et al., 2019, 2025). From this perspective, challenges and contrasts are not necessarily obstacles to learning. Rather, they may constitute resources for productive engagement with mathematical ideas by directing students' attention towards critical aspects of the content.

From a multimodal perspective (e.g., Kress, 2010), such challenges can be understood in terms of tension within and between modes. Tension refers to a perceived discontinuity or potential contradiction in how mathematical content is communicated through modes and can be seen as the inversion of cohesion. Such tension may arise within a single part or across successive parts when modal resources highlight partially divergent aspects of the same mathematical content, and students are required to actively coordinate these expressions. According to Engebretsen (2012), learning environments require a balance between cohesion and tension: sufficient continuity

to establish meaningful connections, and sufficient contrast to create engagement. However, empirical studies suggest that multimodal tension does not uniformly direct students' attention towards the intended mathematical relations. When students work with textbook tasks characterized by multimodal tension, their focus may shift towards surface features or towards resolving relations within and between modes, rather than towards the mathematical content itself (Österholm et al., [in press](#)). This indicates that while some forms of tension may be productive, others may obscure critical mathematical aspects.

These findings indicate that while tension is necessary for learning, not all forms of tension are equally productive. More fine-grained knowledge is therefore needed about what kinds of interactions within and between modes are made available to students, and how these interactions relate to specific mathematical content. In this study, mathematics textbooks are used as a context for examining such interactions, focusing on the introduction of equality and subtraction, two foundational concepts that rely heavily on the coordination of symbols, language, and visual representations, and whose underlying mathematical structures can be foregrounded in different ways through these modes.

Background

Research in mathematics education shows that combinations and interactions between modes are central to how mathematical meaning is constituted. Modes in a task, or in other mathematical texts and parts of text, may align cohesively, but they can also be contradictory so that one mode (e.g., mathematical symbols) expresses one meaning, while another mode (e.g., writing) expresses another meaning (Leijon & Lindstrand, [2012](#)). Translation between symbolic and pictorial modes can be demanding (Lin et al., [2013](#)), and images can draw attention to less relevant information, leading students to focus on those parts of the text rather than the mathematical content being taught (Sutherland et al., [2001](#)). At the same time, other studies show that students can be more successful on tasks where images are used instead of words to describe the mathematical context (Hoogland et al., [2018](#)). Research has also shown that the mathematical symbolic language, especially in relation to natural language, creates complexity in the teaching of mathematics (Johnsson & Österholm, [2019](#)), and that mathematical texts including mathematical symbols require a different type of reading than mathematical texts without these kinds of symbols (Jellis, [2008](#); Österholm, [2006, 2008](#)).

Regarding combinations and interactions between modes, most existing research in mathematics education has focused on aspects of similarities and continuity between modes and their necessity for meaning making, both in analysis of learning materials as a whole (e.g., Bezemer & Kress, [2010](#)) and within specific tasks (e.g., Dyrvold, [2020](#); Nugroho, [2010](#)). At the same time, contrasts between modes have been argued to force active engagement with the content, and both similarities and contrasts are considered necessary for productive learning (Engebretsen, [2012](#)). However, empirical studies focusing on contrasts and discontinuity for analysis are rare in mathematics education (Dyrvold, [2020](#); Engebretsen, [2012](#); Johansson et al., [in press](#)). There is

therefore a strong need for research that investigates the nature of interactions within and between different modes and how such interactions are constituted in relation to mathematical content.

Mathematics textbooks constitute a particularly relevant context for examining such interactions. Textbooks play a pivotal role in planning and preparing mathematics lessons in countries all over the world, particularly for introduction of new content and students' homework (Glasnović Gracin & Jukić Matić, 2016; Jukić Matić & Glasnović Gracin, 2016; Lepik et al., 2015; Leshota & Adler, 2018; Pansell & Björklund Boistrup, 2018; Pepin et al., 2013; Qi et al., 2018). However, empirical studies show that textbooks in mathematics are challenging for students to engage with and that this is related to the combination of different modes (Dyrvold, 2016; Norberg, 2020). Students may therefore need explicit support in how to interact with textbooks, particularly in how to handle different modes and their combinations (Norberg, 2019). Despite the central role of textbooks and the multimodal complexity they present, research explicitly focusing on multimodal aspects of textbooks remains scarce (Norberg, 2020). While Fan et al. (2013) call more broadly for process-oriented research on how textbooks are produced, we consider the multimodal design of textbooks to be a related area of inquiry that can provide knowledge relevant to textbook development. More specifically, we argue that there is a need for greater knowledge about the nature of multimodal interactions. Empirical studies of students' work with textbook also indicate that multimodal tension does not necessarily direct attention toward mathematically relevant relations (Österholm et al., *in press*). Instead, students' attention may shift toward surface features or toward resolving relations between modes rather than toward the underlying mathematical content. These findings suggest that while certain forms of tension may be productive, multimodal tension may also shift attention away from explicit mathematical aspects, depending on how students establish relations between modes and content.

In summary, in order to understand how multimodal interactions may shape students' opportunities to discern mathematical relations, more nuanced knowledge is required about what kinds of interactions within and between modes are made available, and how these interactions relate to specific mathematical content. This study expands the field's understanding of multimodal interactions in textbooks by focusing the analysis on issues of multimodal tension.

Multimodal analysis must be conducted in relation to a subject content (Danielsson & Selander, 2014), in the present study this content is delimited to equality and subtraction. Equality and subtraction were selected because both concepts are foundational for later mathematical learning and require coordination across modes to discern critical mathematical relationships. In equality, students need to coordinate symbolic, visual, and verbal representations to develop a relational rather than operational understanding. In subtraction, students need to coordinate representations that foreground different subtraction structures, such as change and comparison situations. Furthermore, both concepts can be understood in qualitatively different ways, and these distinctions are central to how the mathematical content is communicated through modes. The following paragraphs therefore discuss key perspectives on equality and subtraction that are relevant for the present study's multimodal focus.

Understanding equality is regarded fundamental for algebraic thinking (e.g., Behr et al., 1980; Blanton et al., 2015; Falkner et al., 1999; Kieran, 1981; Knuth et al., 2006; McNeil et al., 2006). The symbolic representation for equality is the equal sign and the ubiquitous presence of the equal sign at all levels of mathematics highlights its importance (Knuth et al., 2006). Young children often show an early understanding of “equal” when comparing physical objects (Falkner et al., 1999). However, difficulties arise as soon as symbolic representations, including the equal sign, are introduced. A well-documented misconception among students is to interpret the equal sign operationally, as a signal to produce an answer, rather than relationally. For instance, students often write 12 as the missing number in the Eq. $8+4=+5$, treating “=” as “makes” rather than expressing a relationship between equivalent expressions (Falkner et al., 1999). A central distinction in the literature is therefore between an *operational* and a *relational* understanding of the equal sign (e.g., Blanton et al., 2015; Knuth et al., 2006). A relational understanding implies recognizing the equal sign as representing equivalence between two quantities, which would lead students to identify the missing number above as 7. It is furthermore discussed that to develop a relational understanding, students must be provided with opportunities to meet equalities/equations like the one above (e.g., Blanton et al., 2015; Falkner et al., 1999; Powell, 2015). However, the mere occurrence of the above equality is not enough as students still can find the missing number by relying only on computation (e.g., adding 8 and 4 and then subtract 5). Instead, students must be supported in noticing structural relationships between the expressions (e.g., that 5 is one more than 4, thus the missing number needs to be one less than 8) (Carpenter et al., 2005). Using a balance scale is highlighted as a way to allow students to investigate equality and sameness (Powell, 2015; Sung et al., 2025), and studies have also shown that introducing balance as a model for solving equations help develop students’ relational understanding (Warren & Copper, 2005). Furthermore, introducing the equal sign together with other relational signs, such as $<$ and $>$, and focus on comparisons have shown to promote relational understanding (Hattikudur & Alibali, 2010). By coordinating concrete, visual, and symbolic representations, and by varying which aspects remain constant and which change, instruction can help students discern the underlying relational structure of equality. For the purposes of this study, both operational and relational manifestations of equality and the equal sign are relevant for the analytical focus.

Beyond mathematics education, research suggests that people systematically privilege additive over subtractive transformations (Adams et al., 2021), indicating that subtraction may be conceptually less salient. This raises questions about how educational materials introduce subtraction and what multimodal resources are used to foreground its underlying mathematical structure. Within mathematics education, understanding subtraction requires students to discern different subtraction situations and the quantitative relationships they embody. Research commonly distinguishes between two major categories: change-take-from and compare (Fuson, 1992). These categories capture the fundamental conceptual structure of subtraction. Children often begin developing understanding of subtraction through concrete scenarios, such as removing or comparing objects. However, students may struggle to

distinguish between different subtraction situations, even when surface features of tasks differ (Bently, 2012; Ekdahl et al., 2025). This suggests that flexible subtraction reasoning requires understanding underlying quantitative relationships rather than relying on single computational strategies. This view aligns with relational perspectives on arithmetic, where understanding the relationships between quantities is considered foundational, while arithmetic operations are regarded as ways of expressing or reconstructing these relationships (Polotskaia et al., 2018, 2022). From such a perspective, different representations may foreground different aspects of the underlying quantitative relations, making subtraction particularly relevant for a multimodal analysis. A key distinction in the literature concerns the underlying structure of subtraction situations. *Change-take-from* involves a unary active operation where a quantity is transformed (e.g., “five cats in the room, one walks away, how many are left?”). *Compare* involves a binary static operation, comparing two quantities without altering them (e.g., “five cats in one room vs. two cats in another room, what is the difference?”). Fuson (1992) also identifies *equalize* as a third type, involving a binary active operation (a combination of change-take-from and compare), but this structure requires explicit cues in the task since equalization takes place only in the calculation process itself, and is therefore less central in many analyses. These distinctions derive from previous research in the area. Describe subtraction in terms of separating, comparison, part-part-whole, and joining, where several categories correspond to Fuson’s. Similarly identify change, compare, equalize, and combine. More recent work (Bartolini Bussi et al., 2013) refines these structures into nine types, organized under change, combine, and compare, explicitly highlighting relationships between addition and subtraction. Research also highlights the importance of representational support in helping students develop flexible strategies for subtraction. Studies show that coordinating and systematically varying representations (e.g., the number line and schematics drawings), help students discern critical aspects of subtraction, such as the relation between the two quantities in a comparison or the transformation in a change situation (Ekdahl et al., 2025). Consequently, understanding subtraction involves more than mastering a computational procedure; it requires coordinating representations that foreground different quantitative relationships and subtraction structures.

Theory

This study adopts a multimodal perspective on cohesion and tension. Modes refer to different resources for communication (e.g., images, sound, mathematical symbols, writing). As discussed in the introduction, across several traditions in mathematics education, learning is understood as requiring active engagement with mathematical content that does not immediately make sense. This idea is reflected in constructs such as productive struggle (Hiebert & Grouws, 2007), productive failure and creative mathematical reasoning (Lithner, 2017), all of which emphasize the importance of students actively making sense of mathematical relationships rather than merely reproducing procedures. Common across these perspectives is the assumption that learning opportunities emerge when students encounter challenges that are accessible

yet not immediately resolved. From a multimodal perspective, such challenges may arise through the relations established within and between modes. Therefore, when combining modes, issues of cohesion and tension are always present (Engebretsen, 2012). Cohesion involves similarity, proximity, and continuity to create bonds between semiotically heterogeneous textual elements and, for example, to relate textual content to previous knowledge. Aspects of tension are the inversion, that is, contrast, distance and discontinuity, which force students to actively interact with a text. A balance between cohesion and tension is considered a condition for learning, and “tensions between the parts must be deep and sharp enough to be challenging, though not deeper and sharper than what is possible to bridge” (Engebretsen, 2012, p. 150). This implies that whether tension is experienced depends on the relation between the reader and the text. These theoretical notions of modal affordance, cohesion, and tension allow for an intrinsic analysis of multimodal interactions and their potential to constitute challenging situations promoting productive struggle and conceptual understanding.

In this study, we focus on the aspect of tension, since it is through contrasts and discontinuities that multimodal interactions may create situations requiring active reflection. Such situations can be understood as constituting the multimodal conditions for productive struggle, as discussed in the background. While cohesion enables integration, tension introduces the complexity that may foreground or obscure particular mathematical relations. Furthermore, tension can arise in the interaction within and between different modes, for example when images, symbols and words foreground partially divergent aspects of the same mathematical content, such as exemplified in Fig. 5 below; and within a single mode, for example within images or words, when internal elements point toward different interpretations, such as exemplified in Fig. 4 below. In this study, multimodal tension is therefore used as an umbrella term for contrasts and discontinuities in how mathematical content is communicated through modes, whether constituted within a single mode or between modes. Tensions may also be constituted at different levels of a textbook text. Some tensions emerge across successive parts, such as between an example and a subsequent task, while others emerge within a single part. Distinguishing between these levels is important because tensions established across parts may create different opportunities for coordinating mathematical meaning than tensions established within a single textbook element. In the present study, this distinction serves as an analytical lens for examining how tensions relate to mathematical content.

Tension can be present with respect to different interactional dimensions of the text: performativity (i.e., levels of objectivity of the represented world), materiality (i.e., perceptual and aesthetic aspects), and semantics (i.e., the actual meaning of the text) (Engebretsen, 2012). In this study, we focus only on the semantic dimension. The semantic dimension includes the correlations between persons, objects, places, and events that are referred to when using different modes, as well as what statements about the world are put forward in the different modes and how easy these are to integrate in a cohesive text universe (Engebretsen, 2012). In the context of this study, concerning modes in mathematics textbooks, statements about the world refer to statements of the mathematical content in focus. Furthermore, Engebretsen (2012) highlights the importance of considering both denotative and connotative level of

semiosis. In a mathematics textbook context, this will include both direct references to the mathematical content, and indirect references such as different explanation models (e.g., subtraction as take-away) or metaphors (e.g., balance scale for equality) for the specific mathematical content.

Purpose & RQ

The purpose of this study is to extend the research field's understanding of multimodal tensions and how such tensions relate to conceptual mathematical content. This is done by adopting a multimodal perspective that uses the concept of tension to empirically describe how interactions within and between modes are established in mathematics education. The study is delimited to interactions within and between modes in the introductory parts concerning equality and subtraction in mathematics textbooks. The following research questions guide the study:

- What types of multimodal tension can be identified in the introductory presentations of equality and subtraction in mathematics textbooks?
- How do the identified tensions relate to the mathematical content in focus?

Method

To address the research questions, an exploratory qualitative analysis of introductory textbook chapters on equality and subtraction was conducted.

Data

Equality and subtraction are concepts introduced in the first grade in the Swedish school system, students aged 7–8 years. Thus, textbooks are limited to those available for grade one. Textbooks are defined as printed learning materials, with or without digital components, that correspond to the national curricula and are published by an established/professional publisher (The Swedish Education Act, SFS 2010:800). However, there is no national control or regulation of approved textbooks in Sweden. Nor is there any record of sales statistics from publishers of textbooks for Swedish schools. To increase opportunities to identify a broad range of multimodal interactions in relation to the mathematical concepts, three different textbook series from well-established publishers with different instructional designs were included in the analyses, *Favorit matematik* (Ristola et al., 2018), *Mitt i Prick* (Mårtensson & Öhman, 2016), and *Rik matematik* (Ryve et al., 2021). Commonly, Swedish mathematics textbooks consist of student books and related teacher guides. Both *Favorit matematik* and *Mitt i Prick* could be considered as traditional textbooks in the sense that there is one student book for each semester, divided into a number of chapters, which include examples and tasks with explicit instructions for students to solve, and can be used in mathematics education without addition of the related teacher guide. Whereas for *Rik*

matematik (relatively new on the market) the main book is the teacher guide, divided into chapters, with the student books only including some of the tasks suggested to work on and no examples.

The ways in which the mathematical content was introduced varied across the different textbooks. To find the chapter where the concepts were introduced, each textbook series was searched for the first time any words, images, or symbols showed an explicit focus on each concept. For example, “equal to”, “equality”, “equal sign”, an image of a balance scale, or “=”; and “subtraction”, “take away”, “compare”, images with crossed over parts, or “-”. For Favorit matematik and Mitt i prickt, the search departed from the student books, and then corresponding passages in the teacher guides were identified. For Rik matematik, the search departed from the teacher guide and then corresponding sections in the student books were identified. If the particular symbol for each concept (i.e., “=” and “-”) was not present in the identified chapter, the first succeeding chapter introducing the symbol was also included, since mathematical symbols are central in mathematics and for our multimodal focus.

The analysis included all parts of the identified chapters, such as definitions, examples, and tasks, as well as other resources referred to in the textbook, such as, video clips or animated presentation. Parts not foregrounding any conceptual aspects, such as only focusing on algorithms for calculation without relation to any of the two subtraction situations, or parts focusing on practicing writing the symbols (“=”, “-”), were excluded. Furthermore, since our focus is on the multimodal interactions *students* may encounter, only those parts of the teacher guides that directly relate to activities intended for classroom teaching were included. Parts directed solely at the teacher, such as explanations of the mathematical content or arguments for particular instructional choices, were excluded. Additionally, all three textbook series included digital components that came with the purchase of a student book, concerning both tasks for students to solve and examples for teachers to show to students. Those digital components that were explicitly suggested in the student textbooks or in the teacher guides to include in teaching were also included in the analysis.

The decision about which parts would be included in the analysis was made jointly by all three authors. In total, 82 pages and an additional 13 instances of web resources were included in the analysis. Each textbook was independently analysed by two of the three authors, with the author pairings varying across the material. The purpose of this parallel analysis was to broaden the analytical perspective and increase the likelihood that relevant instances of cohesion and tension would be identified. The authors subsequently compared their analyses, including instances identified by only one author and cases interpreted differently. These cases were discussed in relation to the focus of the study. An instance was retained only if the identified modal interaction was relevant to how equality or subtraction was presented. For example, visual distinctions serving an organisational function only, such as the use of pink for example boxes and blue for tasks, were not treated as instances of tension. Decisions about inclusion and about whether an instance represented cohesion or tension were reached through discussion and consensus.

Analysis process

1. In the first step, the authors examined each part of the chapter to identify the modes used to present equality or subtraction, and how the modes interacted. Cohesion and tension were identified both at a micro-level, referring to interactions within a single textbook element (e.g., within a specific task or worked example), and at a macro-level, referring to interactions between successive textbook elements (e.g., between different tasks, or between a picture and a task) (exemplified below). Candidate instances were retained only when the identified modal interaction was relevant to the presentation of the focal mathematical content.
2. The second step focused on the retained instances of tension. The first author compiled the analytic descriptions from the initial analyses and led a bottom-up comparative analysis. Instances were compared with regard to similarities and differences in how tension was constituted in relation to the focal mathematical content and were grouped into provisional thematic categories. Although the analytical level of each instance had been identified in Step 1, the categories were not constructed as level-specific in advance; a category could therefore, in principle, have included both micro- and macro-level instances. However, when the provisional categories were examined in relation to the analytical level of their constituent instances, each category was found to comprise instances occurring consistently at one level. All authors then reviewed the proposed categories, discussed their scope and boundaries, and reconsidered the placement of individual instances. Ambiguous cases were examined in relation to the surrounding textbook context and the mathematical content in focus. Category descriptions and assignments were revised through discussion until a shared interpretation was reached. The resulting categories and their distinguishing characteristics are presented in the Results section.
3. In the third step, all authors jointly interpreted how the identified categories of tension related to the mathematical content in focus, that is, equality and subtraction. This involved comparing the individual instances within and across categories to identify recurring patterns in whether the mathematical content remained semantically coherent, became fragmented, or was ambiguously anchored.

Here is an example of the analysis process. If parts of the included chapter in the analysis consist of an activity to work on in whole class (Fig. 1, top) and a task for students to work on their own (Fig. 1, bottom), the modes in each part are noted. Identified modes in the whole class activity are words (both written and spoken according to instruction in the teacher guide), icons (coloured dots) in a matrix, numerals (5) with an empty box between, and a coloured number line with numbers 0–10, and eventually the symbol = (Fig. 1, top). And in the individual student task, identified modes are words (written), images (fingers), icons (coloured cells in matrix and coloured dots in a circle), numerals and words, numerals, and number lines in the example (right).

Then cohesion and tension are identified within each part (micro-level) and between different parts (macro-level). To be considered as cohesion between modes,

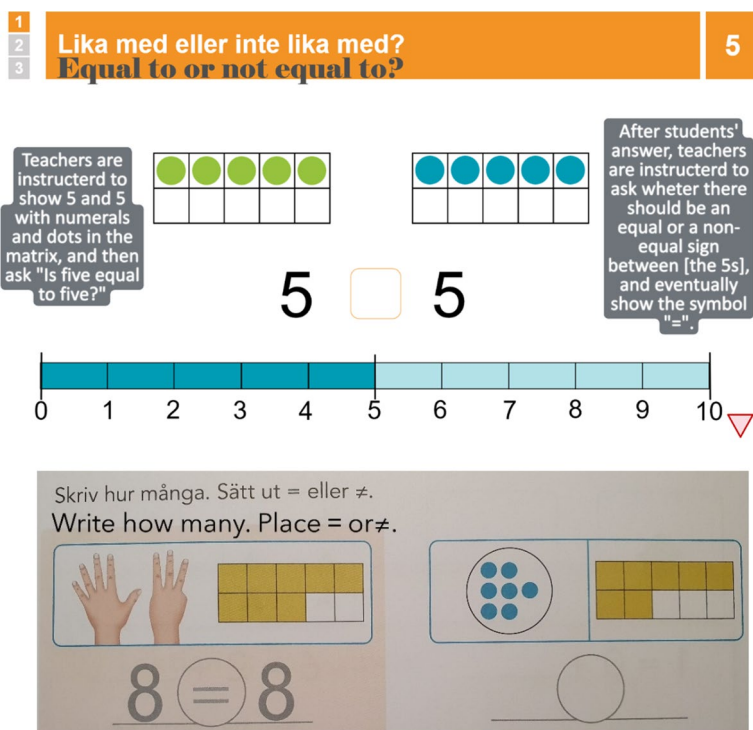


Fig. 1 Parts of an analysed chapter. Note. Translated from Swedish. Top: a slide from a recommended PowerPoint, with instructions added in grey boxes; bottom: corresponding student-book pages (Rik Matematik 1 A; teacher guide, p. 107, ppt; student book, Chap. 2, p. 19. Published with permission)

the different modes must communicate consistent information. For example, there are five green dots in the left matrix, which is consistent with the information by the left symbol 5, and in similar way the information by the five blue dots to the right is consistent with the right symbol 5 (Fig. 1, top). Furthermore, the use of the equal sign between both 5-symbols is consistent with the use of the equal sign between both 8-symbols in the student task (Fig. 1, bottom). To be considered as multimodal tension, there had to be a contrast or discontinuity in how the mathematical content was communicated through modal resources. However, the configuration still had to be interpretable in relation to the same mathematical content. Cases where no such relation could reasonably be established were regarded as incoherence rather than tension and were therefore excluded from the categorisation of tensions. For example, the five dots in the left and right matrixes have different colours (Fig. 1, top), and in the student task (Fig. 1, bottom) there are eight fingers and eight filled boxes in the left square, which could be considered not directly consistent with the two identical 8-symbols with the equal sign in between. This tension in how the image mode is used may contribute to making the mathematical meaning of the equal sign explicit, that is, that it is only the number of objects that matter, not colours or type of objects.

Results

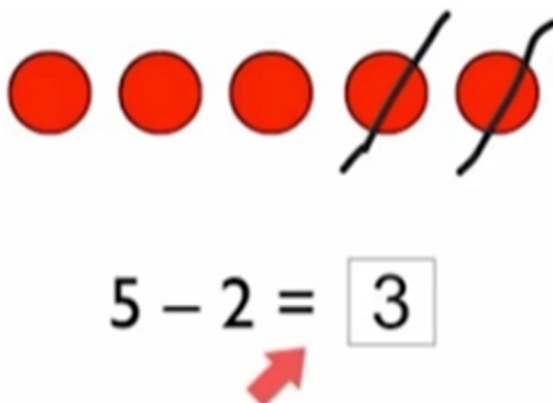
The analysis identified five qualitatively different types of multimodal tension in the introductory presentations of equality and subtraction. These categories answer RQ1 by describing the types of tension that occur in the analysed material: *Changes within the same mode*, *Changes in modal combinations*, *Expected use of previously introduced modes*, *Different aspects highlighted by different modes*, and *Unclear role of a mode*. At the same time, the analysis of each category addresses RQ2 by examining how these tensions relate to the mathematical content in focus. The identified tensions occur at two different analytical levels. Three categories arise at a macro level, where tension unfolds across successive tasks or parts of the text, while two categories occur at a micro level, where tension emerges within a single task or part. This distinction proved analytically important, as tensions at these two levels tend to establish different kinds of relations between modes and mathematical content. The five categories are presented below. The section ends with a summary table to provide an overview of the results.

Changes within the same mode (macro level)

This category captures forms of tension that arise when the same mode is used in different ways across successive parts to communicate an aspect of subtraction or equality. The tension occurs between parts (macro level), as the representational form within a single mode shifts over time. For example, in a sequence of tasks, subtraction is sometimes illustrated by crossing over images of items (Fig. 2) and sometimes by images of objects that disappear (Fig. 3). In both cases the image mode is used, but the internal logic differs as in one case the original set remains visible but is marked, while in the other case the set is visually transformed. This creates tension because a coherent interpretation requires recognising that the same underlying mathematical idea can be illustrated in different ways. This may foreground subtraction as take-away by explicitly showing a difference in the number of objects in the set.

Another example concerns equality tasks where students place the equal sign between sets containing the same number of objects. In earlier tasks, the objects are

Fig. 2 Illustration of subtraction as crossing over. Note. From an animated introductory video to the chapter (Favorit Matematik 1 A, Chap. 2. Published with permission)





$$3 - 1 = 2$$

Fig. 3 Illustration of subtraction as objects disappearing. Note. AI-generated illustration of the same type of modal interaction in the animated introductory video to the chapter represented in Fig. 2; the original is copyrighted

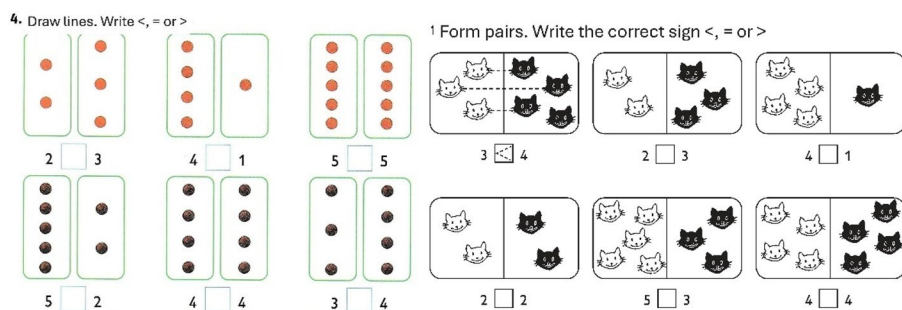


Fig. 4 Illustration of equality as related to the number of objects, not to the appearance of the objects. Note. Translated from Swedish. Left: student-book task; right: complimentary practice-sheet task (Favorit Matematik 1 A, student book, p. 36; practice sheet, Chap. 1, 8a. Published with permission)

visually identical (Fig. 4, left), while later tasks include images of objects of different colours (Fig. 4, right). Although the visual appearance of the objects changes, the number remains constant. This highlights that it is only the number of objects that are important, not what kind of objects.

Across these examples, the mode remains constant (image), while the way subtraction or equality is communicated through that mode changes. Although surface features vary (e.g., crossing over vs. disappearing objects, identical vs. different objects), the mathematical content remains the same. Subtraction is still communicated as take-away, and equality as equivalence in number. The tension thus foregrounds the invariance of the mathematical content despite variation in how it is communicated within the same mode. Since this variation unfolds across successive tasks, the tension occurs at a macro level and highlights the stability of the mathematical concept over time.

Changes in modal combinations (macro level)

This category concerns tension that emerges when the combination of modes used to communicate the mathematical content changes between successive parts. For exam-

ple, subtraction as take-away may first be communicated through a coordinated use of words (“there are 3 birds and 1 flies away”), images (one of the three birds flying from the branch), and symbols ($3 - 1$) (Fig. 5, left). In a succeeding part, however, only words (“1 bird flies away”) and symbols ($2 - 1$) communicate the subtraction, while the image shows both birds sitting on the branch rather than one flying away (Fig. 5, right). This creates tension because an aspect of the mathematical situation that was previously communicated through multiple modes is no longer supported by the same modal configuration. As a result, the relation between modes and mathematical content is reconfigured across tasks.

A similar pattern occurs in relation to aspects of equality, where lines initially support comparison between sets of objects but are absent in succeeding comparisons (Fig. 6). In such cases, modes that previously supported the interpretation of equality are removed or altered.

Through changes in modal combinations, the way subtraction or equality is communicated shifts across parts. Certain aspects of the mathematical content that were previously supported by several modes (e.g., both words and images) may later be communicated by fewer modes, or in different ways. At the macro level, this reconfiguration can make specific aspects of the content more or less explicit, thereby altering how strongly the mathematical relations are foregrounded.

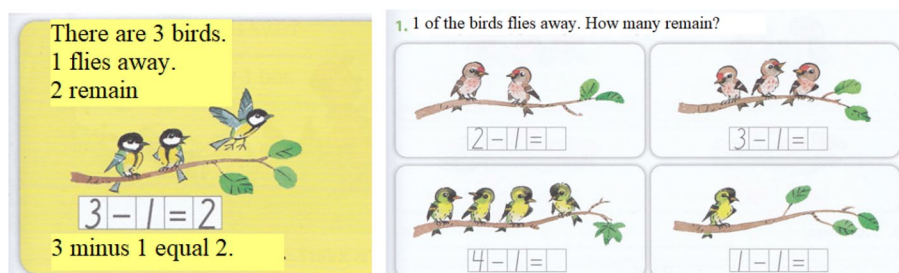


Fig. 5 Illustration of changes in the use of images (birds) in relation to subtraction. Note. Translated from Swedish. On the left is the instruction preceding the task on the right (Mitt i Prick 1 A, student book, p. 51. Published with permission)

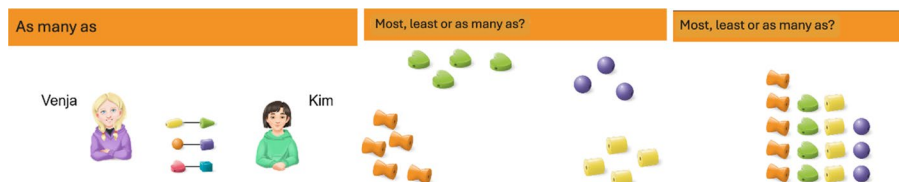


Fig. 6 Illustration of changes in the use of lines in relation to equality. Note. Translated from Swedish. Slides from a PowerPoint teachers are recommended to use in the first lecture in relation to “comparing numbers” (Rik Matematik 1 A, teacher guide, p. 80–81, ppt. Published with permission)

Expected use of previously introduced modes (macro level)

This category captures tensions across successive parts when a particular use of a mode is explicitly established in an earlier example but only implicitly expected in a subsequent task. For example, a preceding example may partly communicate a subtraction situation symbolically. In the following task, students are expected to write the full symbolic expression for the situation themselves (Fig. 7).

Another example concerns equality, where earlier images in the example illustrate equal numbers of objects on both sides of the equal sign. In the succeeding task, students are expected to complete the images so that the number of objects becomes equal on both sides (Fig. 8).

This type of tension positions the mathematical content as dependent on previously established ways of communicating through particular modes. Rather than re-establishing how subtraction or equality is constituted in the task at hand, students are



Fig. 7 Illustration of when symbols are expected to be used by students in relation to subtraction. Note. Translated from Swedish. Subsequent tasks in the student book (Mitt i Prick, student book, p. 51–52. Published with permission)

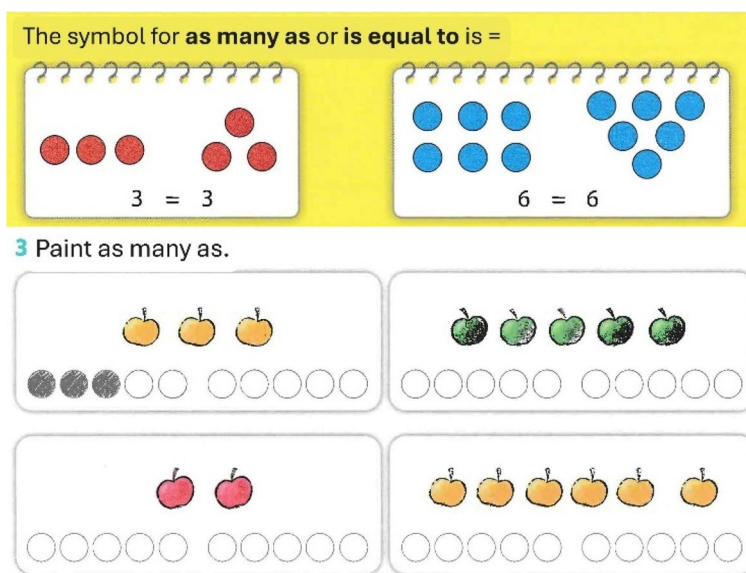


Fig. 8 Illustration of when images are expected to be used by students in relation to equality. Note. Translated from Swedish. At the top, the instructions, and at the bottom, student tasks (Mitt i Prick 1 A, student book, p. 9–10. Published with permission)

expected to reproduce earlier modal patterns. At the macro level, this may foreground similarity in how modes are used, rather than directing attention to the quantitative or relational properties that constitute the mathematical content itself.

Different aspects highlighted by different modes (micro level)

This category describes situations where different modes within the same part foreground different aspects of the mathematical content. For example, when words communicate subtraction as both take-away and comparison, but the accompanying images and symbolic expressions only illustrate subtraction as take-away (Fig. 9). In this case, the written explanation introduces multiple interpretations of subtraction, whereas the visual and symbolic modes emphasise only one of these interpretations.

Another example is when words communicate “the equal sign”, but the symbols show both the equal sign and the non-equal sign, and both equality and non-equality is communicated by images (Fig. 10).

As a result, different aspects of subtraction or equality are foregrounded simultaneously within the same part, but not in a coordinated way. For example, subtraction may be communicated as both take-away and comparison in words, while images and symbols primarily support take-away. At the micro level, such configurations fragment the relation between modes and mathematical content, potentially making it unclear which aspect of the concept is intended to be central.

“Aha, but what does that minus mean?” Sally continued.
 “You surely remember the plus sign. It means addition.
 This minus sign means subtraction—so you take
 something away, or you compare and out the
 difference between two numbers,” Charlie explained.

“Yes, but anyway, whenever something is taken away,
 something is eaten up, something disappears, you
 compare, or even if someone flies away, then it’s about
 subtraction and you need the minus sign,” Charlie
 explained.

Sally suddenly remembered the four seagulls. She
 thought: “So if you have, for example, four seagulls, one
 of them flies away and then there are three seagulls
 left—that’s written as $4 - 1 = 3$ in math language.”

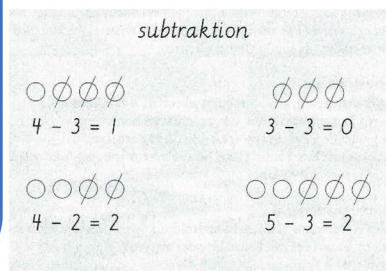
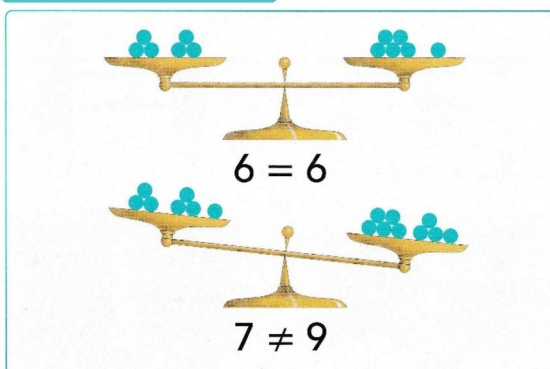


Fig. 9 Illustration of when words communicate subtraction as both take-away and comparison, while interaction of images and symbols relates to subtraction as take-away. Note. Story translated from Swedish. Left: part of an introductory story for listening or teacher reading; right: pictures and symbols to be reproduced on the whiteboard by the teacher (Favorit Matematik 1 A, teacher guide, p. 63. Published with permission)

Fig. 10 Illustration of when symbols and images communicate both equality and non-equality, while words relate to equality. Note. Translated from Swedish. (Rik Matematik 1 A, student book, Chap. 2, p. 17. Published with permission)

Lesson 6: The equal sign



Unclear role of a mode (micro level)

This category concerns situations where one or more modes are present, but their relation to the mathematical content is not clearly communicated. For example, subtraction as take-away may be communicated through images of animals, green dots, words, and symbols (Fig. 11). At the same time, it also included images of five leaves divided into two sets (three and two). These leaves could potentially be interpreted mathematically (e.g. three leaves relating to the symbol 3, or subtraction as comparison). However, the role of the leaves in relation to the subtraction is unclear.

Another example is when all present modes in a task, except the number line, relates to equality (incl. non-equality, Fig. 12), and it is not clear what mathematical content the number line communicates in relation to equality.

In these situations, one mode does not clearly contribute to communicating subtraction or equality. Since its mathematical function is not made explicit, this creates ambiguous relations between modes and content. At the micro level, such ambiguity may obscure which aspects of the mathematical idea are intended to be in focus.

Fig. 11 Illustration of when images (the leaves) can relate to the mathematical content, but probably should not. Note. Because the original image is copyrighted, this is an AI-generated illustration showing the same type of interaction between modes that was present in the introductory example in a student book



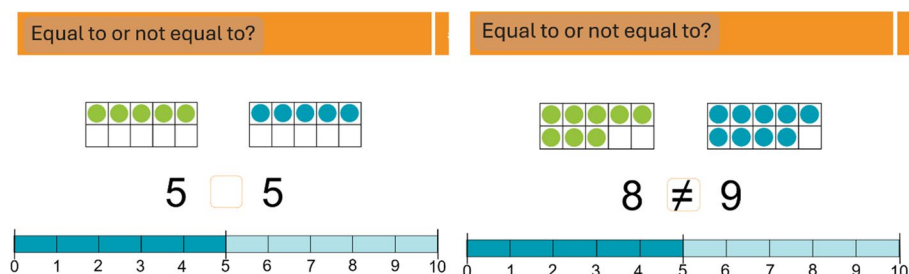


Fig. 12 Illustration of when the relation between one mode (the number line) and the mathematical content (equality) is unclear. Note. Translated from Swedish. Whole-class PowerPoint sequence in which students select a sign between the numbers before the teacher shows the correct sign (Rik Matematik 1 A, teacher guide, p. 107, ppt. Published with permission)

Summary

The analysis identified five qualitatively different types of multimodal tension in textbook introductions of equality and subtraction. As shown in Table 1, the majority of the identified instances occurred at the macro level. Although the categories could in principle have comprised instances from both analytical levels, each category was found to occur consistently at either the macro or the micro level. Comparison across the analysed instances showed that macro-level tensions tended to vary how mathematical ideas were communicated while maintaining semantic coherence, whereas micro-level tensions more often fragmented or ambiguously anchored the mathematical content within a task. Table 1 summarises the identified categories, the number of instances in each category, the analytical level at which they occurred, and their relation to the mathematical content.

Table 1 Categories of multimodal tension, number of instances, analytical level observed across the instances in each category, and relation to the mathematical content

Type of tension	Number of instances	Analytical level observed across instances (macro, micro)	Relation to mathematical content
Changes within the same mode	26	macro	foregrounding invariance of the mathematical content, despite changes in how it is communicated
Changes in modal combinations	12	macro	reconfigured relation between modes and mathematical content, making different aspects of the content more or less explicit
Expected use of previously introduced modes	20	macro	stable expectation of how the mathematical content is communicated, which may shift attention from the content to modal patterns
Different aspects highlighted by different modes	16	micro	fragmented relation between modes and mathematical content, with certain aspects of the content backgrounded
Unclear role of a mode	12	micro	ambiguous relations between modes and content

Discussion

This study set out to explore what types of multimodal tension are constituted in textbook introductions of equality and subtraction, and how these tensions relate to the mathematical content in focus. The results indicate that multimodal tensions are not inherently productive or problematic; rather, their significance depends on how and where they occur. It is important to emphasize that the analysis was exploratory in nature. The categories of tension were not predetermined but emerged through a bottom-up comparison of identified instances. Although the micro–macro distinction was used as an analytical lens from the first stage of analysis, neither the thematic categories nor their association with a particular level was predetermined. Categories could, in principle, have comprised instances from both levels; however, each emergent category was found to consist consistently of either macro- or micro-level instances. Comparison across the instances in these categories subsequently revealed recurring differences in how tension related to the mathematical content at the two levels. That this distinction proved central for how tensions relate to mathematical content suggests that the location of tension is a theoretically significant dimension in multimodal analyses of mathematics textbooks.

Since the present study concerns textbook design rather than textbook use, the following interpretations refer to potential opportunities and challenges embedded in the analysed multimodal configurations. The analysis does not establish how students notice, interpret, or respond to these tensions in classroom practice. Tensions identified on the macro-level, such as changes within the same mode or changes in modal combinations across parts, appear to have greater potential to contribute to what has been described as productive struggle (Hiebert & Grouws, 2007). When the same mathematical idea is communicated through variation within a mode (e.g., different ways of communicating subtraction as take-away within the image mode), invariant aspects of the mathematical content may become foregrounded precisely through contrast. In these cases, tension establishes a relation in which the mathematical content remains stable while the way it is communicated through a mode changes. Such configurations resonate with research emphasizing the importance of systematic variation and coordination across modes for discerning critical aspects (Duval, 2006; Ekdahl et al., 2025). Importantly, macro-level tensions do not necessarily reside within a single part but unfold across sequences of parts. This temporal dimension may allow students to reinterpret earlier modal configurations in light of later ones, potentially supporting deeper structural understanding. In this sense, macro-level tension may create conditions for productive struggle by making invariant mathematical relations visible through variation in how they are represented.

In contrast, tensions identified on the micro-level, particularly those categorized as “different aspects highlighted by different modes” and “unclear role of a mode”, more frequently established fragmented or ambiguous relations between modes and mathematical content. Here, different modes within the same part foregrounded partially divergent aspects of the content, or introduced elements whose mathematical role was not clearly anchored. Such configurations may increase the cognitive demand of the activity, but not necessarily in ways that support conceptual understanding. Instead, tension at the micro-level may shift students’ attention toward resolving dis-

crepancies between modes or toward surface features of the task, rather than toward the intended mathematical relations. This interpretation aligns with findings by Österholm et al. (*in press*), who show that multimodal tension does not uniformly lead students to attend to mathematically critical aspects. Rather, students may orient toward reconciling modes as objects in themselves, instead of coordinating them as expressions of mathematical structure. From a cohesion-tension perspective, this suggests that the “sharpness” of tension (Engebretsen, 2012) is not solely a matter of contrast per se, but of how clearly the mathematical content is semantically anchored across modes. When anchoring is weak, tension risks becoming disruptive rather than productive.

A third pattern in the results concerns situations in which students are expected to use previously introduced modes without explicit re-establishment of their function. These configurations place the mathematical content in a dependent relation to established modal conventions. While this may support fluency in using symbolic or visual forms, it may also reduce engagement to imitation of modal patterns rather than coordination of mathematical relations (Österholm et al., *in press*). In such cases, students may reproduce symbolic expressions or visual completions without necessarily engaging with the underlying quantitative or relational structure. This also resonates with previous research showing that design features of tasks influence which representations students attend to and whether they engage with underlying mathematical relations or with more immediately available procedures (Norqvist et al., 2019, 2025). Here, tension does not arise from semantic contrast between modes, but from the implicit expectation that students should recognize and replicate a modal configuration. The mathematical demand may thus shift from conceptual coordination to formal similarity.

Taken together, the findings nuance the theoretical claim that contrasts between modes inherently foster active engagement (Engebretsen, 2012). While some degree of tension has been theorised necessary for learning in the sense that complete cohesion offers little challenge, the present results suggest that the location, type, and semantic anchoring of tension are decisive for how it relates to mathematical content. The types of macro-level tensions identified in this study vary how mathematical ideas are communicated while maintaining semantic coherence. In contrast, micro-level tensions identified more often fragment semantic alignment or introduce unclear modal roles that may obscure critical aspects of equality or subtraction, particularly in early introductions of these foundational concepts.

Concerning practical implications, the findings highlight the importance of reflecting on how multimodal tensions are distributed over parts in textbooks and used in the classroom. The predominance of macro-level tensions in the analysed material may reflect an emphasis from textbook authors on progression across parts, where variation over time may provide opportunities for conceptual development. Such progression appears to create conditions in which invariant aspects of equality or subtraction can become visible through contrast. At the same time, the findings suggest that tensions within single parts require particularly careful design. When different modes simultaneously highlight divergent aspects, or when a mode lacks a clear mathematical function, the intended mathematical relations may become less salient.

This raises questions for textbook authors and teachers about how micro-level tensions can be structured so that they remain clearly anchored in the mathematical content. Rather than eliminating tension, the challenge may lie in designing tensions that invite coordination of mathematical relations within a task, not merely across tasks. For textbook designers, this implies attending not only to the modes present within individual tasks, but also to how their mathematical functions are maintained or transformed across successive textbook parts.

In this way, the study contributes to the relatively scarce body of research that focuses explicitly on discontinuity and contrast in multimodal textbook design (Dyrvold, 2020; Engebretsen, 2012; Johansson et al., [in press](#)). Through an exploratory analysis, we have identified distinct types of multimodal tension and demonstrate that these tensions establish qualitatively different relations to specific mathematical content. In particular, the empirically grounded distinction between macro- and micro-level tensions highlights the importance of considering not only whether tension is present, but where and how it is constituted in relation to mathematical ideas. The distinction between macro- and micro-level tensions also points toward a need for further research examining whether and how these levels function differently in students' meaning making. While the present study identified the distinction through analysis of textbook design, future studies could investigate how students and teachers engage with macro- and micro-level tensions in classroom practice. In doing so, the study responds to the identified need for more nuanced analyses of how multimodal interactions relate to specific mathematical content and offers conceptual distinctions that may guide future investigations of multimodal textbook design.

Limitations

The present study analyses tensions constituted in textbook design rather than empirically observed learning processes. While textbooks are known to influence the planning and organization of teaching, and thereby shape students' opportunities to encounter mathematical content, the study does not examine how these tensions are enacted, interpreted, or transformed in classroom interaction. Consequently, the findings concern the learning opportunities embedded in multimodal configurations rather than how these opportunities are taken up in practice.

A related limitation concerns the relation between analytically identified tensions and students' experiences of those tensions. As discussed in the theoretical framework, whether tension is experienced depends on the relation between the reader and the text (Engebretsen, 2012). The tensions identified in this study were therefore analysed from the perspective of researchers with established knowledge of the mathematical content in focus. Students encountering equality or subtraction for the first time may not interpret these multimodal configurations in the same way. Consequently, the identified tensions should be understood as potential opportunities and challenges made available through textbook design rather than direct descriptions of students' experiences. Future studies examining students' interactions with textbook materials need to investigate how such tensions are interpreted in practice.

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Declarations

Conflict of interest No potential conflict of interest to report.

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